

Experiment 2: Common Elements Between Two Arrays

Aim

To write a Python program to find the **common elements between two arrays** and analyze its time complexity.

Algorithm

1. Start.
2. Read two arrays.
3. Convert the second array into a set.
4. Traverse each element of the first array.
5. Check whether the element exists in the set.
6. If it exists, add it to the result.
7. Display the common elements.
8. Stop.

Source Code

```
def common_elements(arr1, arr2):
    set2 = set(arr2)
    result = []

    for element in arr1:
        if element in set2 and element not in result:
            result.append(element)

    return result

arr1 = [1, 2, 3, 4, 5]
arr2 = [3, 4, 5, 6, 7]

result = common_elements(arr1, arr2)

print("Array 1:", arr1)
print("Array 2:", arr2)
print("Common Elements:", result)
```

OUTPUT :

```
Array 1: [1, 2, 3, 4, 5]  
Array 2: [3, 4, 5, 6, 7]  
Common Elements: [3, 4, 5]
```

RESULT :

Thus, the Python program was successfully executed and the common elements between the two given arrays, **3, 4, and 5**, were identified.